

STAT 6560
Graphical Methods
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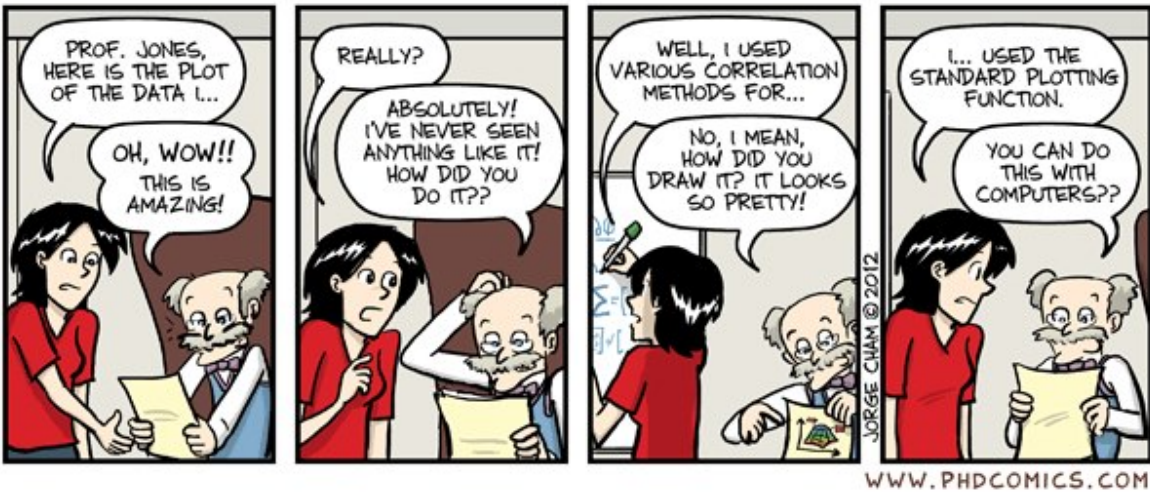


Figure 1: <http://www.phdcomics.com/comics/archive.php?comicid=1541>, Cartoon.

Contents

Acknowledgements	1
1 Introduction: A Couple of Good and Bad Examples	1
1.1 Motivation	1
1.2 Why Graphics ?!?	3
1.3 How to Display Data Badly	4
1.3.1 Don't show much data	5
1.3.2 Show the data inaccurately	10
1.3.3 Obfuscate the data	17
1.4 Bad Graphics are Everywhere — In Space and in Time	33
1.5 Rules for Good Data Displays	44
1.6 Further Reading	48
2 History of Statistical Graphics: Plots, People, and Events	50
2.1 General History	50
2.1.1 Milestones in the History of Data Visualization (According to Friendly)	51
2.2 Selected People	52
2.3 Statistical Graphics and Events in History	65
2.3.1 John Snow and the Cholera Epidemic in London, 1854	65
2.3.2 The Challenger Disaster, 1986	69
2.4 Further Reading	73
3 Graphics Galleries and Sources on the Web	74
4 Color and Cognition	76
4.1 Color-Deficiency and Color-Blindness	76
4.2 Various Color Spaces	81
4.2.1 The HSL and HSV Color Spaces	81

4.2.2	The RGB Color Space	83
4.2.3	The HCL Color Space	85
4.3	Suggestions for Color Selections	86
4.4	Good Color Choices	91
4.4.1	Work by Cindy Brewer and Collaborators	93
4.4.2	Work by Zeileis, Hornik, and Murrell	96
4.5	Change Blindness	97
4.6	Further Reading	100
5	Statistical Maps	101
5.1	Choropleth Maps	101
5.1.1	Choropleth Maps in R	105
5.2	Linked Micromaps	109
5.2.1	Template for LM Plots	110
5.2.2	Micromaps vs. Choropleth Maps	112
5.2.3	Additional Linked Micromap Examples	117
5.2.4	Web-based Applications of LM Plots	123
5.2.5	Linked Micromaps in Java	126
5.2.6	Linked Micromaps in R	127
5.3	Conditioned Micromaps	131
5.4	Comparative Micromaps	134
5.5	Further Reading	137
6	Categorical Plots	138
6.1	Which Plot Type to Choose?	138
6.2	Categorical Plots in R	142
6.2.1	Pie Charts	144
6.2.2	Bar Charts	146
6.2.3	Dot Charts	147

6.2.4	Mosaic Plots	147
6.2.5	Spine Plots and Spinograms	148
6.2.6	Four Fold Plots	148
6.3	Categorical Plots in Mondrian	150
6.3.1	Installation	150
6.3.2	The Titanic Data in Mondrian	151
6.4	Further Reading	153
6.5	R Code and Output	154
6.5.1	Example 1: UCBA admissions	155
6.5.2	Example 2: Titanic	167
6.5.3	Example 3: HairEyeColor	170
7	Univariate Plots	178
7.1	Histograms	178
7.2	Averaged Shifted Histograms	186
7.3	Stem-and-Leaf Plots	189
7.4	Boxplots (or Box-and-Whisker Plots)	190
7.5	Dot Charts for Univariate Data	191
7.6	Kernel Density Plots for Univariate Data (with Rug Plot)	193
7.7	Quantile-Quantile Plots (Q-Q Plots)	195
7.8	Empirical Cumulative Distribution Functions (ECDFs)	198
7.9	Graphics and Small Sample Sizes	200
7.10	Further Reading	204
8	Bivariate Plots	205
8.1	Scatterplots	205
8.2	Hexagon Binning	208
8.3	Bivariate Histograms	209

9	Trivariate Plots	212
9.1	Scatterplot Matrix	212
9.2	3D Scatterplots	213
9.3	Co-Plots	214
9.4	Trivariate Density Estimation	216
10	“Hypervariate” (High-Dimensional) Plots	220
10.1	Scatterplot Matrix (for $n \geq 4$)	220
10.2	Parallel Coordinate Plots	221
10.3	Faces, Star Plots, and other Glyph Representations	223
10.4	Andrews Plots	225
10.5	Data Images	228
10.6	From Tables to Plots	230
11	Interactive and Dynamic Graphics	237
11.1	Software for Interactive and Dynamic Graphics	238
11.1.1	REGARD, MANET, and Mondrian	238
11.1.2	HyperVision, ExplorN, and CrystalVision	241
11.1.3	Data Viewer, XGobi, and GGobi	244
11.2	Concepts of Interactive and Dynamic Graphics	248
11.2.1	Scatterplots and Scatterplot Matrices	248
11.2.2	Brushing and Linked Brushing/Linked Views	249
11.2.3	Focusing, Zooming, Panning, Slicing, Rescaling, and Reformatting	250
11.2.4	Rotations and Projections	252
11.2.5	Grand Tour	252
11.3	Spatial Data Analysis in the Linked ArcView 2.1 and XGobi Environment	254
11.3.1	Basic XGobi Layout	257
11.3.2	Graphical Methods	258
11.3.3	Linking Between More Complicated Objects for Data Analysis . .	265

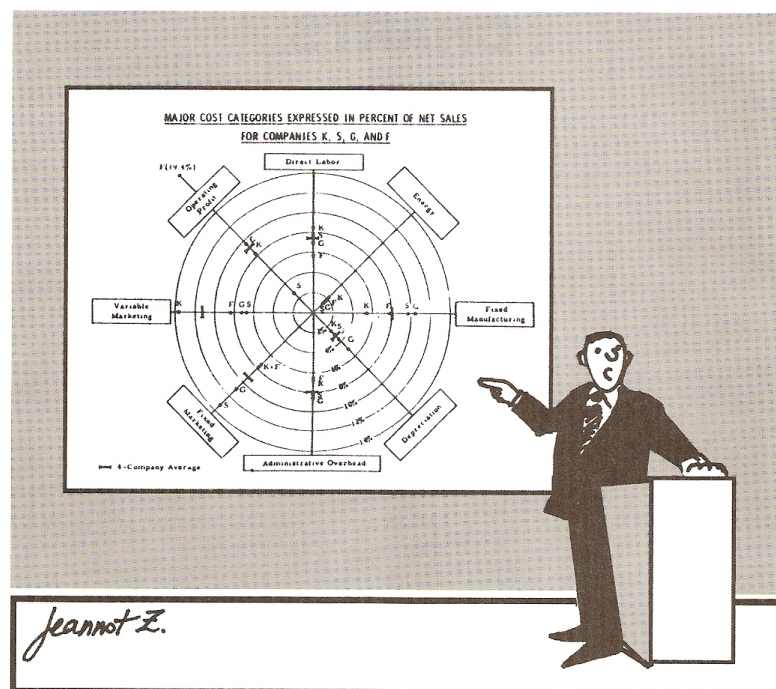
11.4 Interactive and Dynamic Graphics in R	269
11.4.1 R Package <i>rggobi</i>	269
11.4.2 R Package <i>tourr</i>	271
11.4.3 R Package <i>iplots</i>	272
11.4.4 R Package <i>animation</i>	274
Appendix	276
Homework Assignments	277
Homework Assignment 1	1
Homework Assignment 2	1
Homework Assignment 3	1
References	1

Acknowledgements

This course uses some of the course materials provided by Dr. Mike Minnotte (formerly USU, now with the University of North Dakota) as held in the Fall 2006 semester. Additional materials have been taken from other Statistical Graphics courses, such as the ones offered by Dr. Di Cook (Iowa State University: <http://www.public.iastate.edu/~dicook/>) and Dr. Dan Carr (George Mason University: <http://mason.gmu.edu/~dcarr/>). We are likely to include parts from additional web sources that will be specified later on.

Thanks are also due to around 20 students and guests who took Stat 6560 with me since the Spring 2009 semester for their valuable comments that helped to improve and correct these lecture notes.

Jürgen Symanzik, August 26, 2012.



"What do you mean, what does it mean?"

Figure 2: Zelazny (2001), p. x, Cartoon.

5 Statistical Maps

5.1 Choropleth Maps

Symanzik & Carr (2008), p. 270, state:

“Choropleth maps use the color or shading of regions in a map to represent region values. Choropleth maps have proved very popular but have many problems and limitations as indicated by writers such as Robinson et al. (1978), Dent (1993), and Harris (1999). [...]

There are two kinds of choropleth maps, called unclassified and classed. Unclassed maps use a continuous color scale to encode continuous values (statistics). This is problematic because perception of color is relative to neighboring colors and because color has poor perceptual accuracy of extraction in a continuous context. Classed choropleth maps ameliorate this problem and dominate in the literature.

Classed choropleth maps use class intervals to convert continuous estimates into an ordered variable with a few values that can be represented using a few colors. When a few colors are easily discriminated and regions are sufficiently large for color perception, color identification problems are minimal. The color scheme also needs to convey the class ordering based on values. Brewer (1997) and Brewer et al. (1997) provided results evaluating different color schemes in a mapping context. The Web site <http://colorbrewer.org> (see Leslie 2002, for a short description) contains guidance on ordered color schemes and additional issues such as suitable schemes for people with color vision deficiencies and for different media. Perfect examples on how colors should be used in choropleth maps can be found in the 1996 “Atlas of United States Mortality” (Pickle et al. 1996).”

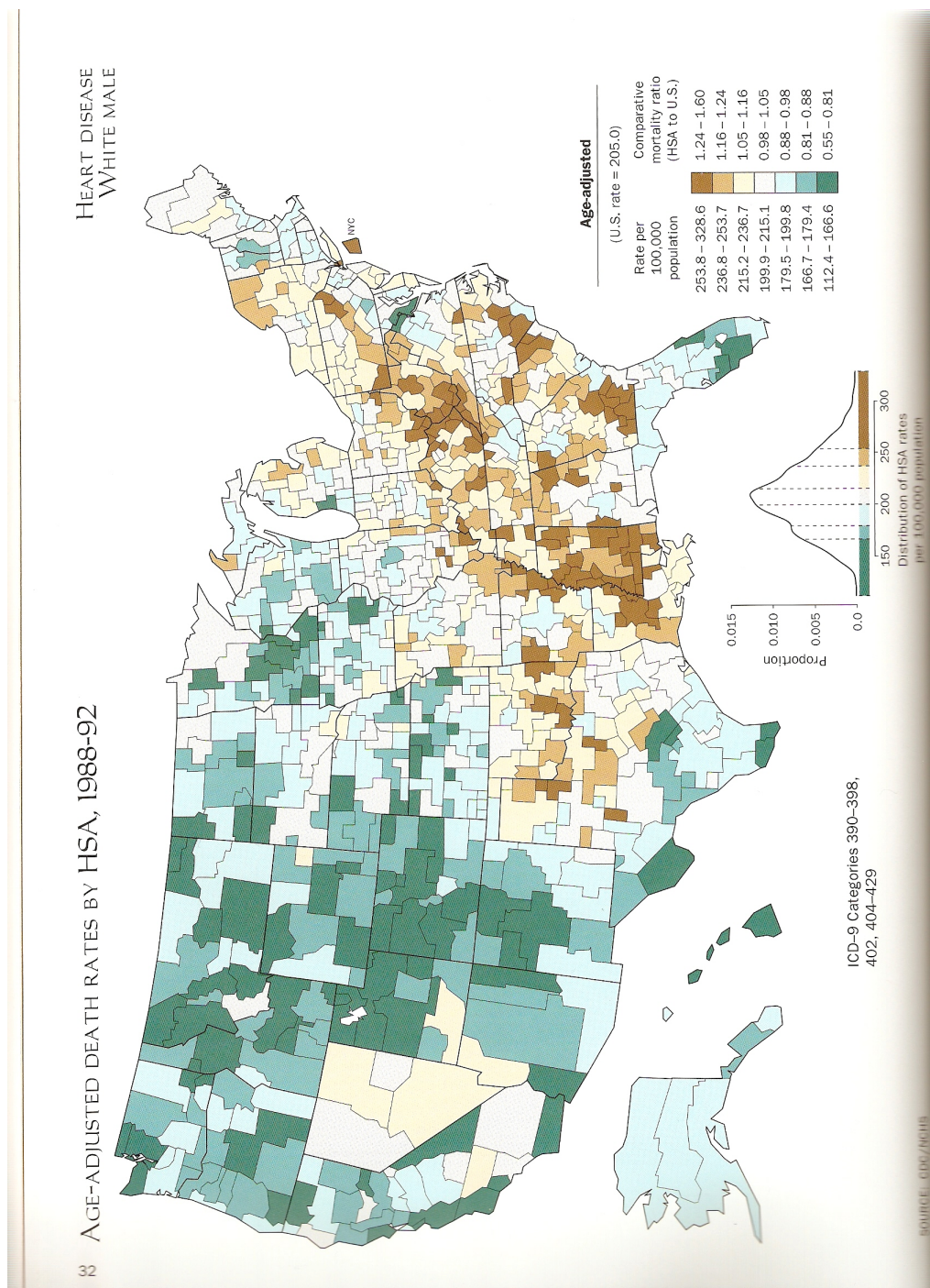


Figure 94: Pickle et al. (1996), p. 32, Figure, showing Heart Disease White Male by Health Service Area (HSA), 1988-92.

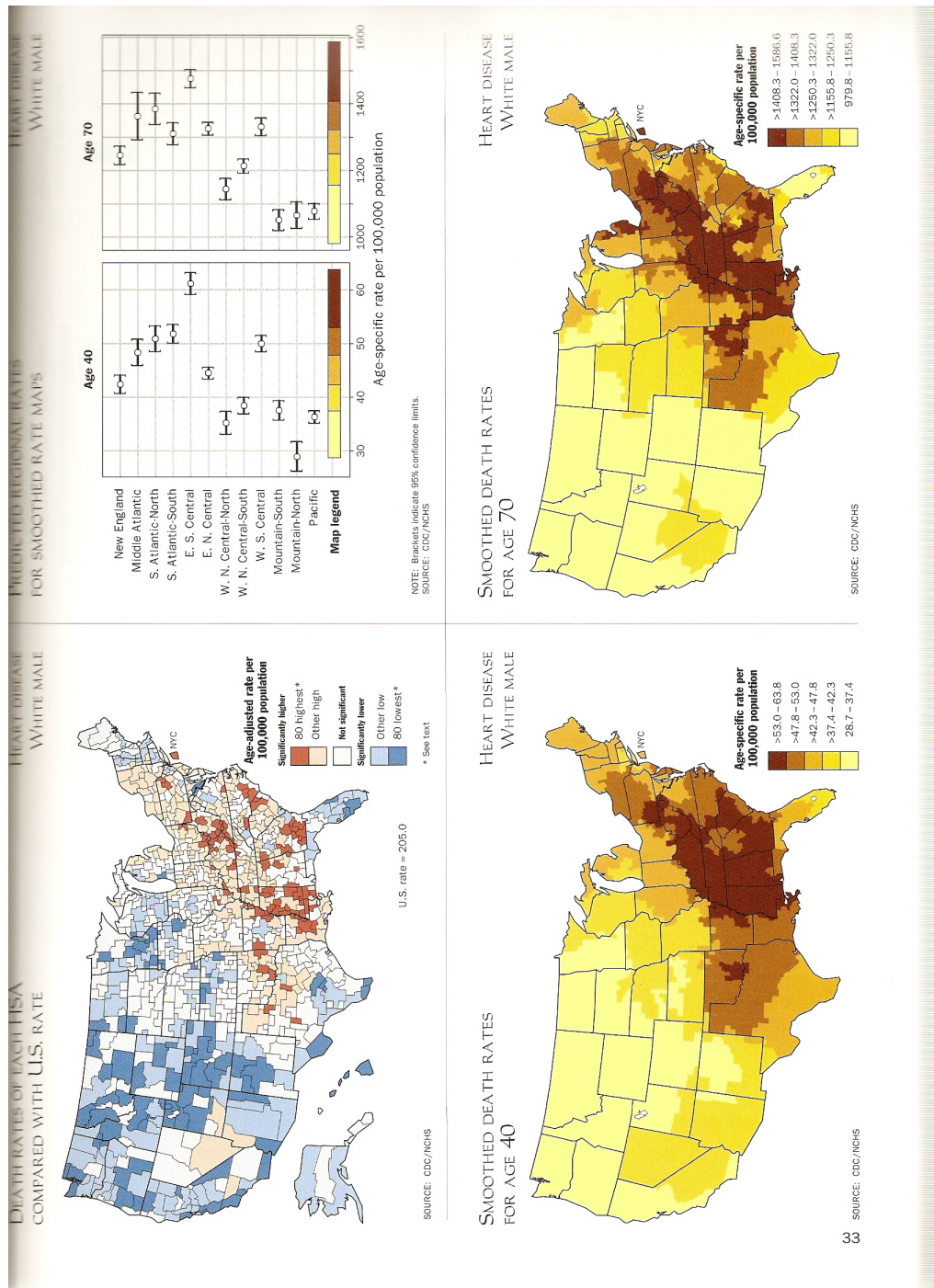


Figure 95: Pickle et al. (1996), p. 33, Additional Figures, showing Heart Disease White Male, 1988–92.

Choropleth Map Construction:

Choropleth Maps are easy to construct on paper. <http://www.teachingideas.co.uk/geography/chlormap.htm> teaches children age 5 to 11 “How to make a Choropleth Map”.

Weather Maps:

We are used to see various new choropleth maps almost daily, in newspapers, on TV, or on the Web. Probably the most popular choropleth maps are weather maps, such as provided by <http://www.usatoday.com/weather/default.htm>. Move your mouse over the various options (Radar, Satellite, Precip, Temps, Fronts) to see various weather-related choropleth maps.

Unfortunately, except for Temps, there are usually no exact values shown on choropleth maps. So, it is difficult to exactly compare different regions or cities.

5.1.1 Choropleth Maps in R

Various R packages support the creation of maps and choropleth maps in R, provide commands to read and write ESRI shapefiles, and link to various Geographic Information Systems (GIS). A more detailed overview can be found at <http://geodacenter.asu.edu/projects/rsp/content/map-packages-cr>.

Example 1:

Basic geographic maps, ranging from a world map to detailed county maps with labels and city names.

```
library(maps)

map() # World (default)

map("state") # US

map("state", c("Utah", "Colorado", "Idaho", "Wyoming", "Montana"))

data(state)
state.region
state.name[state.region == "Northeast"]
map("state", state.name[state.region == "Northeast"])

map("state")
map.axes() # add latitude (North/South) and longitude (East/West)

map("county") # US counties

map("county", "Utah") # Utah counties

map.text("state") # too many labels

map.text("state", state.name[state.region == "South"])

map("county", "Utah")
map.cities() # too many labels
```

```

map("county", "Utah", xlim = c(-113,-111), ylim = c(40,41))
map.cities()
map.axes()
points(c(-112.5, -111.5), c(40.2, 40.8), pch = c(17, 20), col = "red", cex = 1.5)
  ## fill in two additional points

map("state", state.name[state.region == "Northeast"], fill = T, col = 0:8)

library(RColorBrewer)

map("state", state.name[state.region == "Northeast"], fill = T,
  col = brewer.pal(9, "Set3"))

map("state", fill = T, col = brewer.pal(9, "Set3"))

map("state", fill = T, col = brewer.pal(5, "Blues"))

```

Example 2:

Choropleth maps.

```

library(maps)

data(state)

murder <- state.x77[,5]
range(murder)

breaks <- c(1, 4, 7, 10, 13, 16)
m.class <- cut(murder, breaks)
brewer.pal(5, "Blues")
m.col <- brewer.pal(5, "Blues")[m.class]

map.m.col <- m.col[match.map("state", state.name)]
map("state", fill = T, col = map.m.col)
legend("bottomright", legend = levels(m.class), fill = brewer.pal(5, "Blues"))

```

The next four examples incorporate google maps as background into an R graphics window. This can be done with the package *RgoogleMaps*. The main documentation for this package can be found at <http://cran.r-project.org/web/packages/RgoogleMaps/index.html>. The reference manual is located at <http://cran.r-project.org/web/packages/RgoogleMaps/RgoogleMaps.pdf> and additional examples can be found at <http://cran.r-project.org/web/packages/RgoogleMaps/vignettes/RgoogleMaps-intro.pdf>.

The following R packages are required to run the next four examples and should be first installed before running these examples: *RgoogleMaps*, *PBSmapping*, *maptools*, *ReadImages*, and *rgdal*.

The R code below originates from the Web sites above, but has been further modified in some cases. Here is an example how David Kahle, a Ph.D. Candidate in the Department of Statistics, Rice University, has used *RgoogleMaps* in combination with *ggplot2*: <https://github.com/hadley/ggplot2/wiki/Crime-in-Downtown-Houston%2C-Texas-%3A-Combining-ggplot2-and-Google-Maps>. Unfortunately, we cannot reproduce the maps as the data files are not provided.

Example 3:

Google map of Washington, D.C., as background.

```
library(RgoogleMaps)
library(PBSmapping)
library(maptools)

shpFile <- paste(system.file(package = "RgoogleMaps"), "/shapes/bg11_d00.shp", sep = "")

shp = importShapefile(shpFile, projection = "LL")
bb <- qbbox(lat = shp[, "Y"], lon = shp[, "X"])
MyMap <- GetMap.bbox(bb$lonR, bb$latR, destfile = "DC.jpg")
PlotPolysOnStaticMap(MyMap, shp, lwd = .5, col = rgb(0.25, 0.25, 0.25), add = F)

# add some random points to the map
x = runif(100, -0.1, 0.1) + 38.8933115
y = runif(100, -0.1, 0.1) - 77.0146475
PlotOnStaticMap(MyMap, x, y, FUN = points, col = "red", add = T)
```

Example 4:

Google satellite map of lower Manhattan, New York, NY.

```
library(RgoogleMaps)

bb <- qbbox(c(40.702147, 40.711614, 40.718217), c(-74.015794, -74.012318, -73.998284),
  TYPE = "all", margin = list(m=rep(5, 4), TYPE = c("perc", "abs")[1]))
MyMap <- GetMap.bbox(bb$lonR, bb$latR, destfile = "MyTile3.png", maptype = "satellite")

PlotOnStaticMap(MyMap)
```

Example 5:

Google street map of lower Manhattan, New York, NY.

```
library(RgoogleMaps)

bb <- qbbox(c(40.702147, 40.711614, 40.718217), c(-74.015794, -74.012318, -73.998284),
  TYPE = "all", margin = list(m = rep(5, 4), TYPE = c("perc", "abs")[1]))
MyMap <- GetMap.bbox(bb$lonR, bb$latR, destfile = "MyTile3.png")

PlotOnStaticMap(MyMap)
```

Example 6:

Google street map of Washington, D.C., with markers and lines.

```
library(RgoogleMaps)

#Define the markers:
mymarkers <- cbind.data.frame(lat = c(38.898648, 38.889112, 38.880940),
  lon = c(-77.037692, -77.050273, -77.03660), size = c("small", "small", "small"),
  col = c("blue", "green", "red"), char = c("", "", ""))

#get the bounding box:
bb <- qbbox(lat = mymarkers[, "lat"], lon = mymarkers[, "lon"])

#download the map:
MyMap <- GetMap.bbox(bb$lonR, bb$latR, destfile = "DC.png", GRAYSCALE = T,
  markers = mymarkers)

#determine the max zoom, so that all points fit on the plot
# (not necessary in this case):
#zoom <- min(MaxZoom(latrange = bb$latR, lonrange = bb$lonR))

#plot:
library(RColorBrewer)
pal <- brewer.pal(3, "Set2")

tmp <- PlotOnStaticMap(MyMap, lat = mymarkers[, "lat"], lon = mymarkers[, "lon"],
  cex = 1.5, pch = 20, col = pal, add=F)
tmp <- PlotOnStaticMap(MyMap, lat = mymarkers[, "lat"], lon = mymarkers[, "lon"],
  col = c("purple"), add = T, FUN = lines, lwd = 2)
```

5.2 Linked Micromaps

Symanzik & Carr (2008), p. 268, state:

“Over the last decade, researchers have developed many improvements to make statistical graphics more accessible to the general public. These improvements include making statistical summaries more visual and providing more information at the same time. Research in this area involved converting statistical tables into plots (Carr 1994, Carr & Nusser 1995), new ways of displaying geographically referenced data (Carr et al. 1992), and, in particular, the development of linked micromap (LM) plots, often simply called micromaps (Carr & Pierson 1996, Carr et al. 1998, Carr, Olsen, Pierson & Courbois 2000). LM plots, initially called map row plots as well as linked map–attribute graphics, were first presented in a poster session sponsored by the American Statistical Association (ASA) Section on Statistical Graphics at the 1996 Joint Statistical Meetings (Olsen et al. 1996). More details on the history of LM plots and their connection to other research can be found in these early references on micromaps. More recent references on LM plots (Carr, Wallin & Carr 2000, Carr 2001) focused on their use for communicating summary data from health and environmental studies.”

A comprehensive source for micromaps is Carr & Pickle (2010). This book deals with (ordinary) linked micromaps, conditioned micromaps, and comparative micromaps. Three accompanying Web pages at <http://www.crcpress.com/product/isbn/9781420075731>, <http://www.statnetconsulting.com/micromaps.html>, and <http://mason.gmu.edu/~dcarr/Micromaps/> provide access to numerous data sets, shape files, and R code used in the book.

5.2.1 Template for LM Plots

Gebreab et al. (2008), pp. 112–113, state:

“A typical template of a LM plot consists of four key features (Carr & Pierson 1996). Figure 1 shows a hypothetical LM plot. The first feature is **three or more sequence panels in parallel linked by location**. In the hypothetical case, Figure 1 shows five parallel sequences of panels. The first (leftmost) sequence of panels is the micromap panel itself that typically contains small caricatures of map outlines of a region. The caricature map maintains the shape and neighborhood relationship while making the small subregions more visible. The second (from the left) sequence of panels is the label panel that provides the names of the geographical subregions (here, Region 1 through Region 10). The third through the fifth (from the left) sequence of panels display the statistical summaries. These panels may represent many forms of statistical summaries including box-plots, dot-plots (as shown in Figure 1), time series plots, confidence intervals, etc. **Sorting the geographic subregions based on the statistical variable(s) of interest** is the second feature. Sorting improves perception between consecutive panels from the top to the bottom of the display. The third feature is the **partitioning of the regions into perceptual groups of size five or less to allow the viewer’s attention to focus on explicit areas at a time**. The fourth feature is **color and location that links corresponding elements within the parallel sequence panels**, i.e., the color red in the topmost panels relates to the geographic subregion in the northeast of the map, the subregion name (Region 5), and a red dot in each of the three statistical panels. The color red is reused in the next consecutive set of panels for Region 2, but there is no relationship between Region 5 and Region 2 as one might at first assume. Simply, there do not exist enough distinguishable colors to populate an entire display (with, say, 50 different subregions) such that colors have to be reused in different panels.

In the hypothetical Figure 1, the rows are sorted by decreasing values with respect to the statistical panel 2. The statistical data displayed in the statistical panel 1 and 2 show a strong positive association (the correlation r calculated as 0.99), expressed in the almost parallel behavior of the dots and lines representing the values for these two variables. In contrast, the statistical data in panel 3 and 1 (as well as 3 and 2) show a strong negative

association (the correlation r calculated as ± 0.94 for 3 and 1 and as ± 0.92 for 3 and 2). This negative association is seen in the movement of the dots and lines in opposite directions for these variables. Moreover, the data in panel 3 shows an unusual outlier, the value for Region 1. It is this outlier that considerably reduces the almost perfect negative association otherwise present in this data. Just a simple numerical calculation of r might not be able to reveal the influence of a single subregion on the overall relationship.

The map panels of the LM plot in Figure 1 exhibit a strong geographic pattern: Highest occurrences with respect to the statistical panel 1 and 2 can be found in the north and in the east; lowest occurrences can be found in the west and in the south. Additional features of LM plots exist and are described in more details in Symanzik & Carr (2008)."

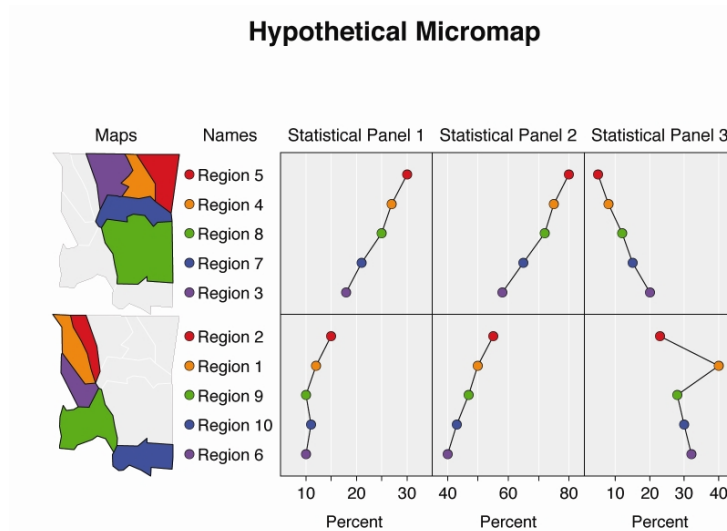


Figure 96: Gebreab et al. (2008), p. 113, Figure 1: Hypothetical LM plot illustrating the main features of such plots: the leftmost sequence of map panels, the second (from the left) sequence of label panels, and the third through the fifth (from the left) sequence of statistical panels.

5.2.2 Micromaps vs. Choropleth Maps

Symanzik & Carr (2008), pp. 270–271, state:

“LM plots often provide a good alternative to displaying statistical information using choropleth maps. Choropleth maps use the color or shading of regions in a map to represent region values. Choropleth maps have proved very popular but have many problems and limitations as indicated by writers such as Robinson et al. (1978), Dent (1993), and Harris (1999). Reviewing these problems helps to indicate why LM plots are a good alternative. [...]

Even with a good color scheme, three key problems remain for classed choropleth maps. **The first problem relates to region area. As suggested above, some map regions can be too small to effectively show color.** Examples include Washington, D.C., on a map of the United States (U.S.) and Luxembourg on a European map. Map caricatures, such as Monmonier’s state visibility map (Monmonier 1993), can address this problem, by enlarging small regions in a way that maintains region identifiability and shows each region touching the actual neighboring regions. Another facet of the area problem is that large areas have a strong visual impact while in many situations, such as in the mapping of mortality rates, the interpretation should be weighted by the region population. Dorling (1995) addressed this problem by constructing cartograms that changed region shapes to make areas proportional to population. Issues related to this approach are region identifiability, and map instability over time as their shapes change with changing populations. Area related problems persist in choropleth maps.

A second key problem is that converting a continuous variable into a variable with a few ordered values results in an immediate loss of information. This loss includes the relative ranks of regions whose distinct values become encoded with the same value. The task of controlling conversion loss has spawned numerous papers about proposing methods for defining class intervals. Today, guidance is available based on usability studies. Brewer & Pickle (2002) indicated that quintile classes (roughly 20% of regions in each class) tend to perform better than other class interval methods when evaluated across three different map reading tasks. Still, the best of the class interval selection approaches loses information.

The third key problem is that it is difficult to show more than one variable in a choropleth map. MacEachren et al. (1995) and MacEachren et al. (1998) were able to clearly communicate values of a second binary variable (and indicator of estimate reliability) by plotting black and white stripped texture on regions with uncertain estimates. However, more general attempts such as using bivariate colors schemes have been less successful (Wainer & Francolini 1980). Thus, choropleth maps are not suitable for showing estimate standard errors and confidence bounds that result from the application of sound statistical sampling or description. It is possible to use more than one map to show additional variables. However, Monmonier (1996, page 154) observed that when plotting choropleth maps side by side it can easily happen that “similarity among large areas can distort visual estimates of correlation by masking significant dissimilarity among small areas.” The change blindness (Palmer 1999, page 538) that occurs as the human eyes jump from one map to another map makes it difficult to become aware of all the differences that exist in multiple choropleth maps and hard to mentally integrate information in a multivariate context.”

Symanzik & Carr (2008), pp. 272–274, provide the following motivational example:

“Fig. 97 shows two variables, the soybean yield and acreage from the 1997 Census of Agriculture for the United States, displayed in two choropleth maps. Five equal size class intervals were chosen for each of the maps. [...]

The two choropleth maps in Fig. 97 indicate that highest yields and highest acreages for soybeans occur in the Midwest. There seems to be some spatial trend, i.e., some steady decrease for both variables from the Midwest to the Southeast. Overall, there appears to be a positive correlation between these two variables since high yields/high acreages and low yields/low acreages seem to appear in the same geographic regions. The correlation coefficient between yield and acreage is only 0.64, suggesting departures from linearity that would be better revealed using scatterplots or LM plots. [...]

In fact, Fig. 98 shows the LM plots of the same two variables as Fig. 97, plus a third statistical panel for the variable production. Data is available for 31 of the 50 U.S. states only. An identical color links all of the descriptors for a region. Successive perceptual groups use the same set of distinct colors. In Fig. 98, the sorting is done (from largest to smallest) by soybean yield in

those 31 U.S. states where soybeans were planted. Here, the points within a panel are connected to guide the viewer's eyes and not to imply that interpolation is permitted. The connecting lines are a design option and can be omitted to avoid controversy or suit personal preferences. The list of 31 U.S. states is not evenly divisible by five. Two perceptual groups at the top and two groups at the bottom contain four states, while three perceptual groups in the middle contain five states. The middle groups require the use of a fifth linking color. Using distinct hues for linking colors works best in full color plots. For grey-level plots, colors need to be distinct in terms of grey-level. Fig. 98 shows different shades of green and is suitable for production as a grey-level plot. Readers new to LM plots sometimes try to compare regions with the same color across the different perceptual groups, but quickly learn the linkage is meaningful only within a perceptual group."

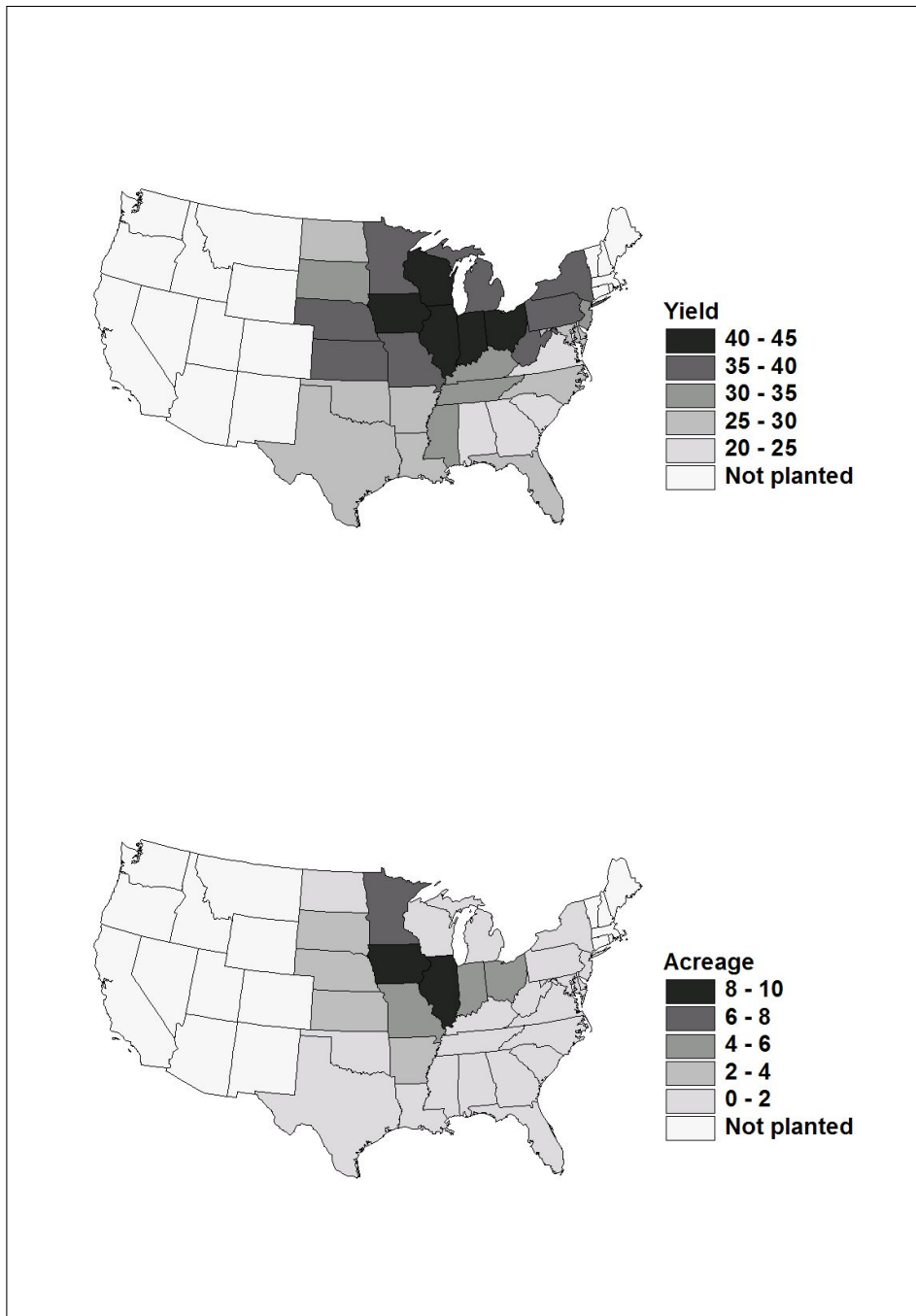


Figure 97: Symanzik & Carr (2008), p. 273, Figure 1.2: Choropleth maps of the 1997 Census of Agriculture, showing the variables soybean yield (in bushels per acre) and acreage (in millions of acres) by state. The data represent the 31 U.S. states where soybeans were planted.

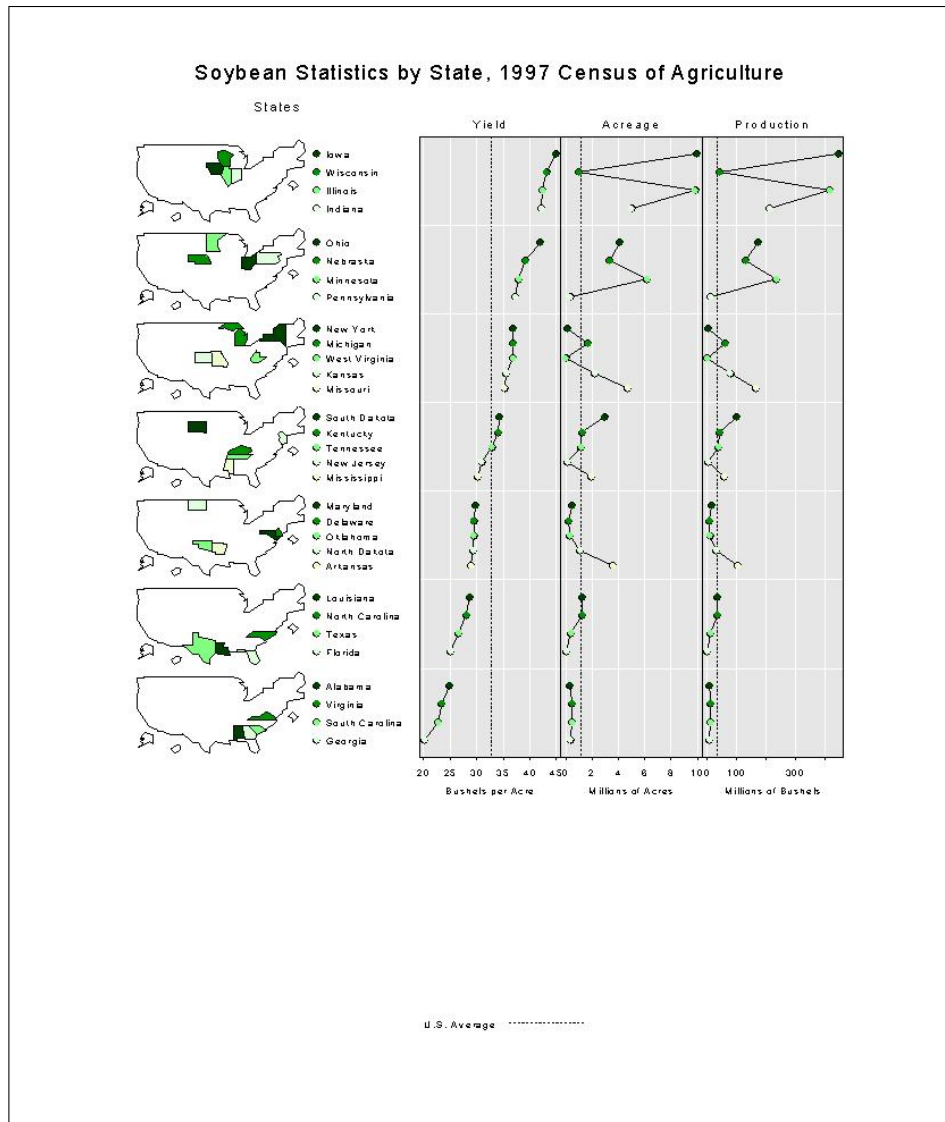


Figure 98: Symanzik & Carr (2008), p. 275, Figure 1.3: LM plots of the 1997 Census of Agriculture, showing soybean yield (in bushels per acre), acreage (in millions of acres), and production (in millions of bushels) by state. The data is sorted by yield and shows the 31 U.S. states where soybeans were planted. The “U.S. Average” represents the median, i.e., the value that splits the data in half such that one half of the states has values below the median and the other half of the states has values above the median. For example, Tennessee is the state with the median yield. This figure has been republished from <http://www.nass.usda.gov/research/gmsoyyap.htm> without any modifications (and ideally should contain much less white space in the lower part).

5.2.3 Additional Linked Micromap Examples

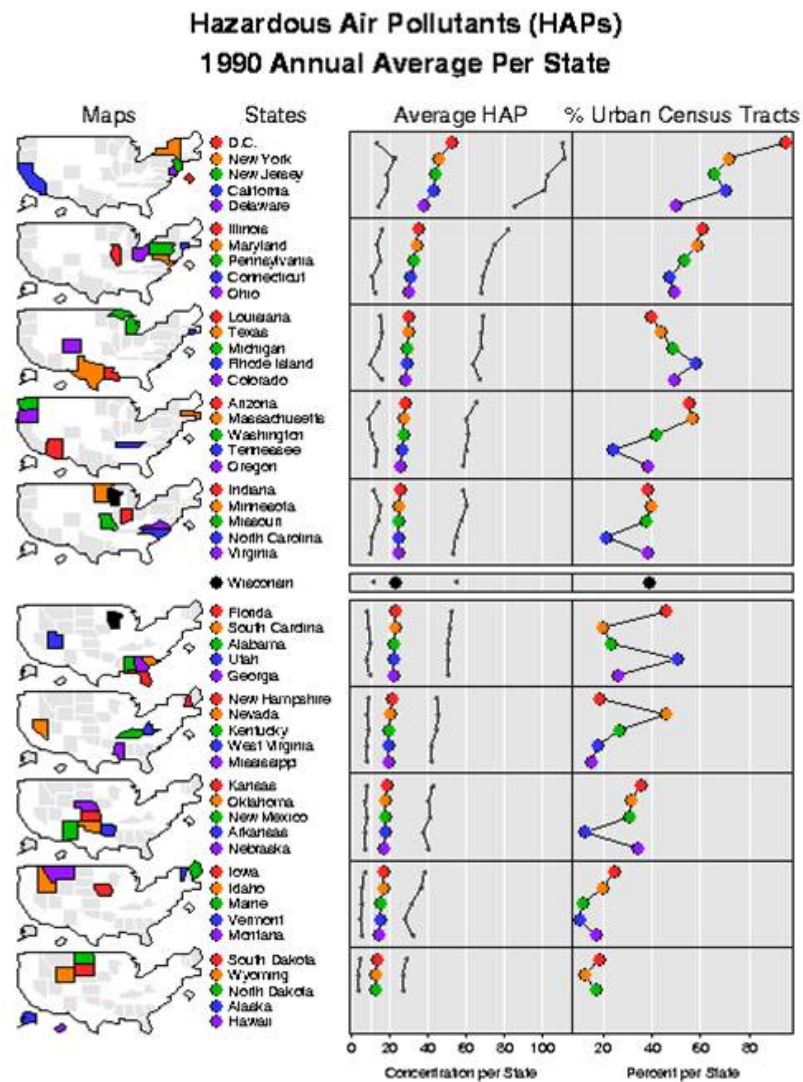


Figure 99: Hazardous Air Pollutants (HAPs) for the US from work with the EPA.

Michigan - Modeled 1990 Air Toxics Concentrations

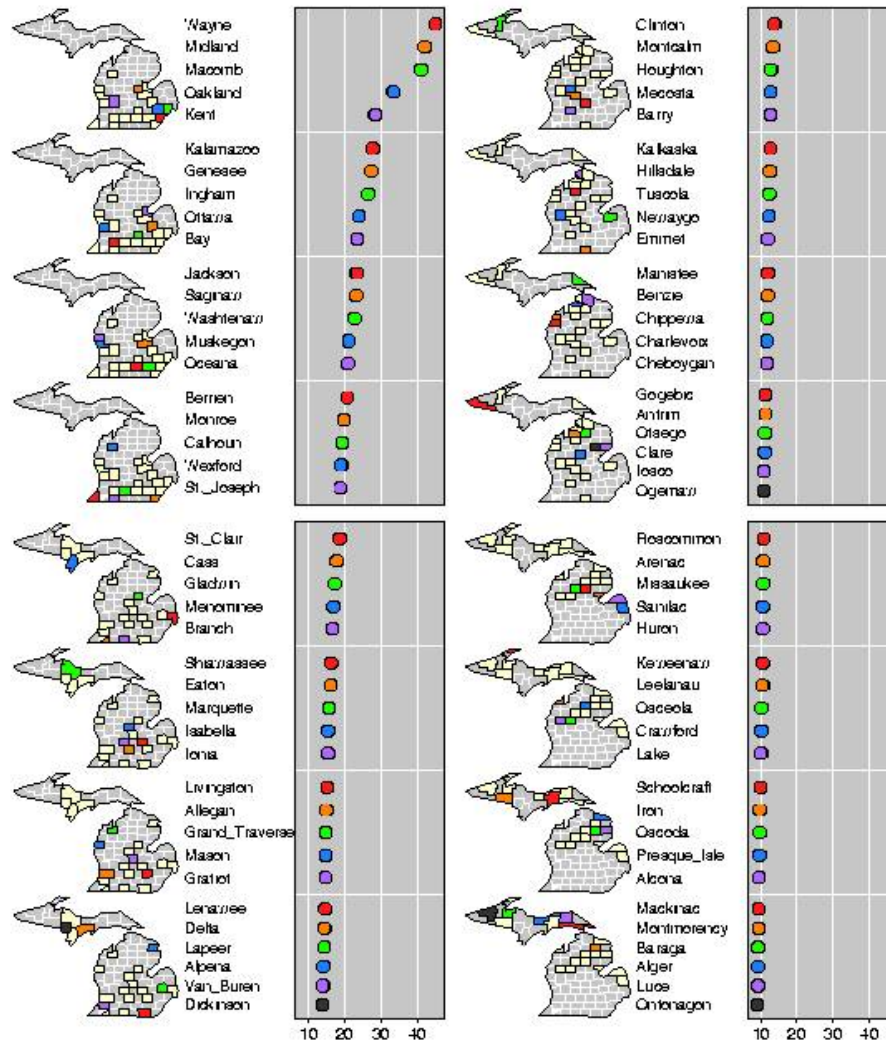


Figure 100: Hazardous Air Pollutants (HAPs) for Michigan from work with the EPA.

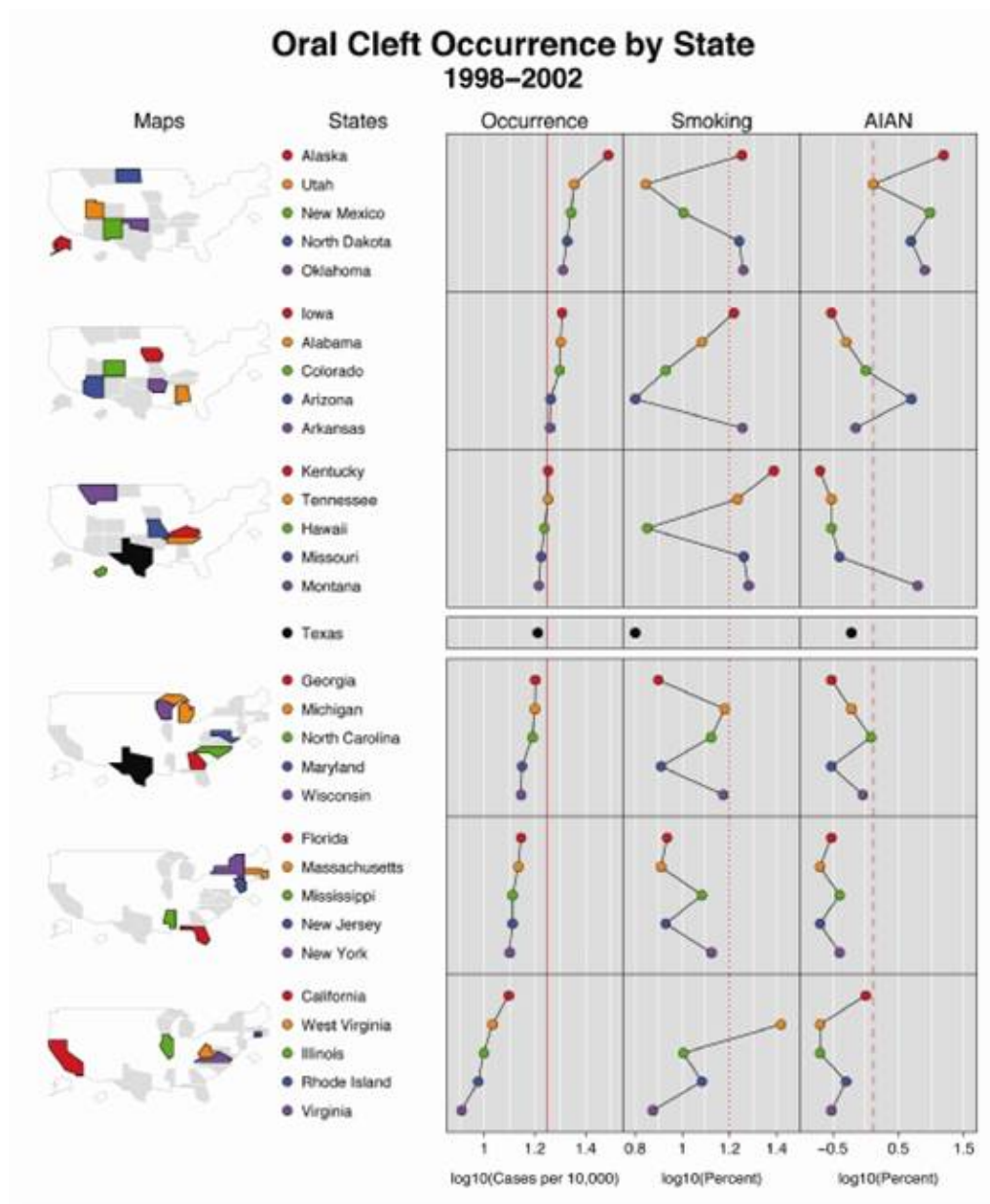


Figure 101: Gebreab et al. (2008), p. 114, Figure 2: LM plot showing oral cleft occurrence by state for the period of 1998–2002. Only oral cleft occurrence for 31 out of the 50 US states was available and displayed here. Smoking rates for California were not available. The red lines show the national average (i.e., mean) of oral cleft occurrence (17.7 per 10,000), smoking rate 16%, and AIAN proportion of 1.3%. Note that Texas had the median oral cleft occurrence among the 31 states for which data were available.

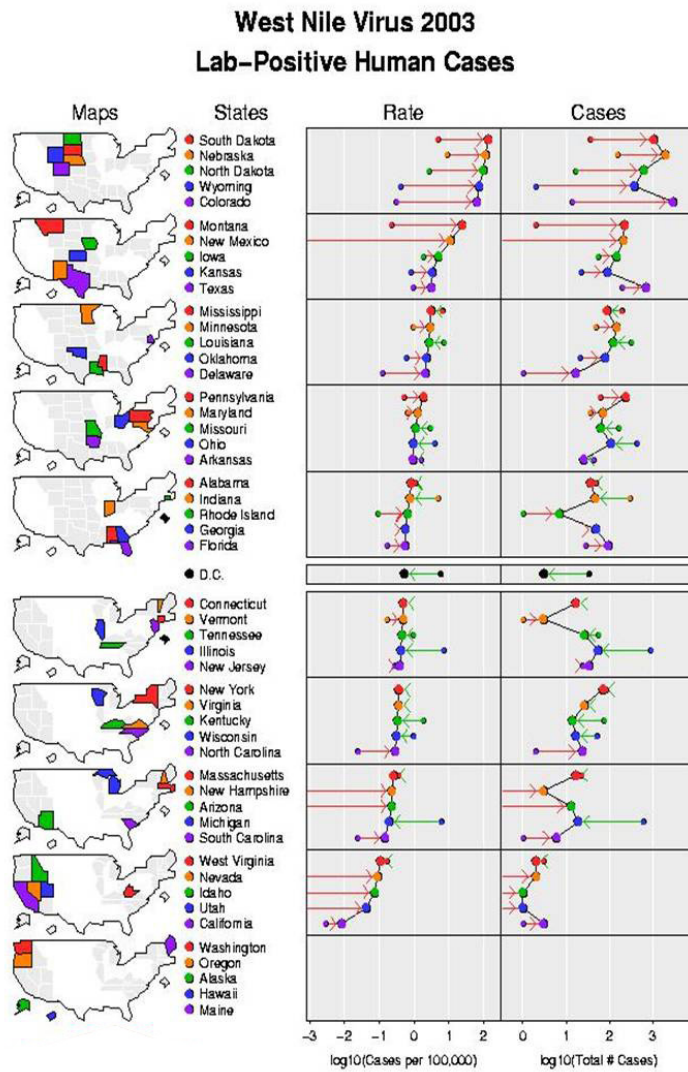


Figure 102: Spread of the West Nile Virus (WNV) across the US, 2002 vs 2003.

Annual CO2 Emissions From Energy Use Units = Tons Per Person

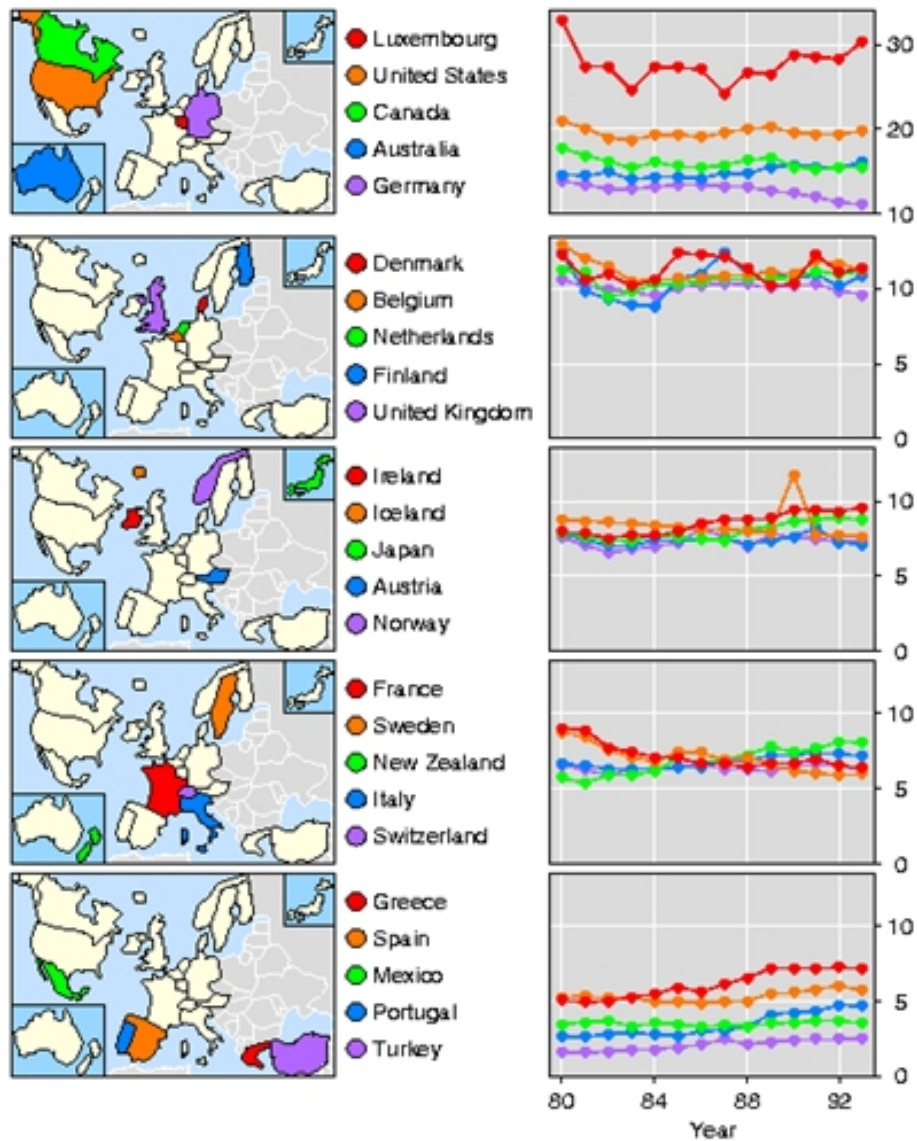


Figure 103: Example from Dan Carr, showing CO2 emissions over time.

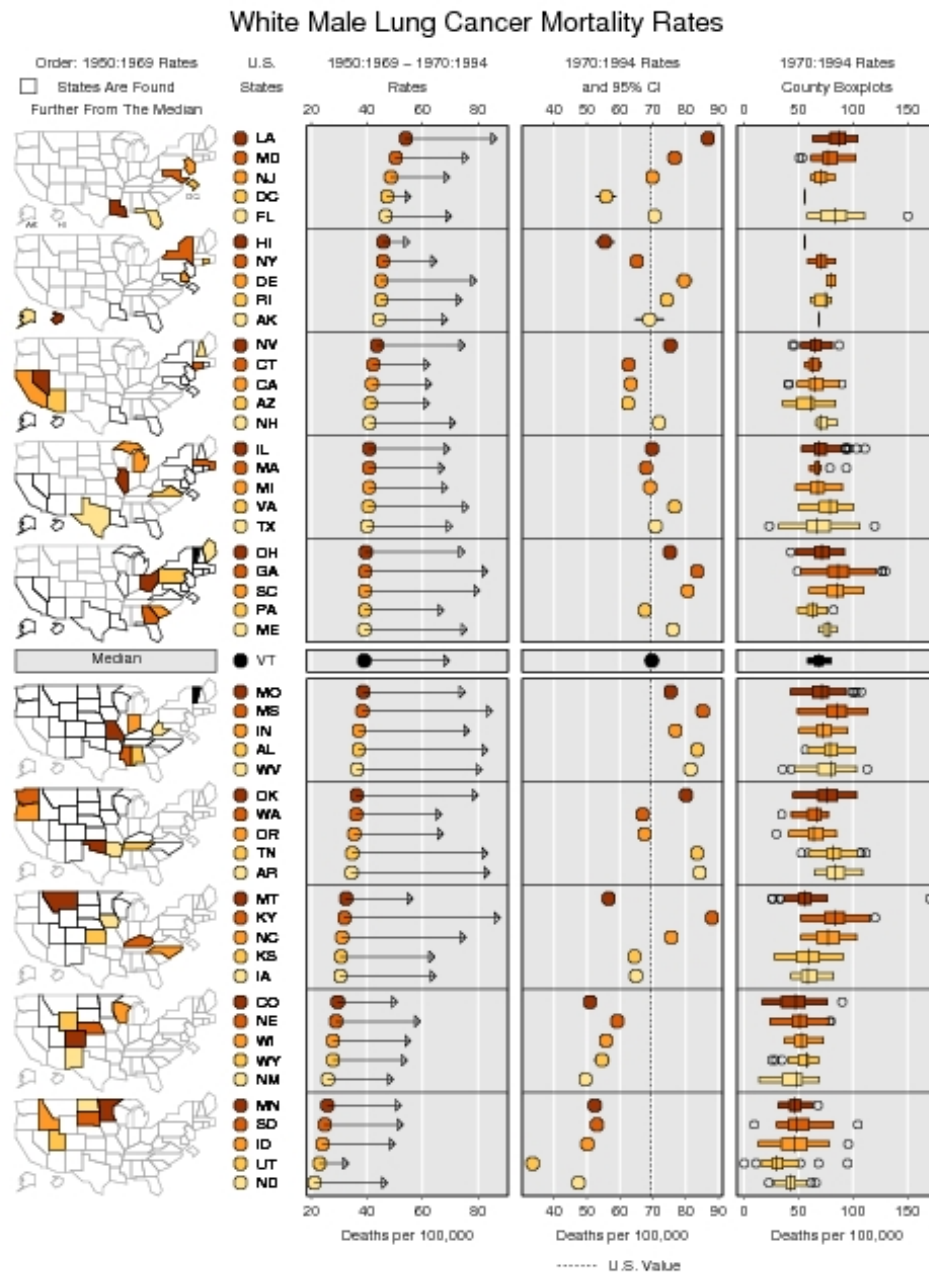


Figure 104: Symanzik & Carr (2008), p. 286, Figure 1.6: LM plots, based on data from the NCI Web page, showing summary values for the years 1950 to 1969 and for the years 1970 to 1994 in the left data panel, rates and 95% confidence intervals in the middle data panel, and boxplots for each of the counties of each state in the right data panel.

5.2.4 Web-based Applications of LM Plots

Symanzik & Carr (2008), pp. 276–282, state:

“Over the last decade, U.S. Federal Agencies and other institutions increasingly focused attention on distributing large amounts of geographically referenced statistical data, either in print or through the Web. The Web-based distribution of data is aimed at replacing printed tabular displays and at providing access to current data quickly. Several approaches have been developed that provide a user-friendly Web-based interface to tabular and graphical displays of Federal data. The user can interactively and dynamically query and sort the data, compare different geographic regions, and look at the data at different spatial resolutions, e.g., at the state or the county level. Carr & Olsen (1996) provide examples on the visual appearance of patterns in data when properly sorted.

The direction of LM plots development shifted from static LM plots towards interactive micromap displays for the Web. Work done for the EPA CEP Web site (Symanzik, Wong, Wang, Carr, Woodruff & Axelrad 2000) was the first in this regard. This project was soon followed by Web-based examples of micromaps produced by the USDA–NASS such as in Fig. 98.

The Digital Government (dg.o) initiative (<http://www.diggov.org>)[‡] is a major research initiative funded by the National Science Foundation (NSF) and several Federal Agencies such as the EPA, the USDA–NASS, the U.S. Census Bureau, the NCI, the U.S. Bureau of Labor Statistics (BLS), etc. This initiative addresses multiple aspects related to Federal data such as visualization, access, disclosure, security, etc. One of the proposals funded under dg.o was the Digital Government Quality Graphics (DGQG) project that included the development of LM plots (<http://www.geovista.psu.edu/grants/dg-qg/index.html>).

In the reminder of this section, we look at four main applications of interactive Web-based LM plots, three of them on Federal Web sites. A short overview of interactive micromaps, as well as a micromap of the “Places” data (Boyer & Savageau 1981), can be found in Symanzik (2004). However, additional details are given in this section.

[‡]Archived versions of these Web pages can be found at http://web.archive.org/web/*/http://www.diggov.org.

Micromaps at the EPA CEP Web Site

The idea of using micromaps on the Web was first considered for the EPA CEP Web site (previously accessible at <http://www.epa.gov/CumulativeExposure/>).[§] Initially, the EPA wanted to provide fast and convenient Web-based access to its hazardous air pollutant (HAP) data for 1990. In this data set, concentrations of 148 air pollutants were estimated for each of the 60,803 U.S. census tracts in the 48 contiguous U.S. states (Rosenbaum et al. 1999). The EPA Web site was designed to allow the user to easily move through the data set to find information on different air pollutants at different geographic locations and at different levels of geographic resolution (e.g., state, county, census tract) via interactive tables and micromaps. Unfortunately, no part of the interactive CEP Web site was ever published due to concerns that the 1990 data was outdated at the intended release date in 1998. Only a static version of the CEP Web site without tables and micromaps was accessible for several years. More details on the work related to the planned interactive CEP Web site can be found in Symanzik, Wong, Wang, Carr, Woodruff & Axelrad (2000), Symanzik, Axelrad, Carr, Wang, Wong & Woodruff (1999), Symanzik, Carr, Axelrad, Wang, Wong & Woodruff (1999).

Micromaps at the USDA–NASS Web Site

The USDA–NASS Research and Development Division released a Web site (<http://www.nass.usda.gov/research/sumpant.htm>) in September 1999 that uses interactive micromaps to display data from the 1997 Census of Agriculture. The USDA–NASS Web site displays acreage, production, and yield of harvested cropland for corn, soybeans, wheat, hay, and cotton. [...]

Micromaps at the NCI Web Site

The NCI released the State Cancer Profiles Web site in April 2003 that provides interactive access to its cancer data via micromaps. This Web site is Java-based and creates micromaps “on the fly”. Wang et al. (2002) and Carr

[§]Archived versions of these Web pages can be found at http://web.archive.org/web/*/http://www.epa.gov/CumulativeExposure/.

et al. (2002) provide more details on the design of the NCI Web site that is accessible at <http://www.statecancerprofiles.cancer.gov/micromaps>. [...]

Micromaps at Utah State University

Micromaps and other graphical displays were found to be very useful for the display and analysis of the geographic spread of the West Nile Virus (WNV) and other diseases (Symanzik et al. 2003) across the U.S. For this reason, researchers at Utah State University (USU) obtained the NCI Java micromap code and adapted it for the display of WNV data (Chapala 2005). Similar to the NCI micromap application, a user can now select among WNV infection rates and counts, and death rates and counts, starting with the WNV data for the U.S. for the 2002 season. A drill-down into U.S. counties is possible given that data at the county-level is available. New features of the USU Web site include the plotting of the data for two years side-by-side in one panel and additional sorting criteria such as sorting from the highest increase over no change to highest decrease in two selected years. The USU WNV micromap Web site can be accessed at <http://webcat.gis.usu.edu:8080/index.html>. [...]"

5.2.5 Linked Micromaps in Java

The National Cancer Institute (NCI) states at <http://gis.cancer.gov/tools/micromaps/>:

“Linked Micromaps is a graphing program written in Java. It allows users to view multiple variables interactively and compare statistics across regions (states, counties, registries, hospitals) as well as across time. It supports six types of graph:

- bar graphs;
- box plots;
- raw data tables;
- point graphs;
- point graphs with arrow; and
- point graphs with confidence intervals.

In order to use Linked Micromaps, you must have Java installed on your PC. Your input files must be in a delimited (such as Comma-Separated Values [CSV]) or fixed-width text format.”

5.2.6 Linked Micromaps in R

Linked micromaps were initially implemented in S-Plus (and not in R). While higher-level functionality is similar, if not identical between both software packages, the internal implementations are rather different. Due to such internal differences, it took many years before the appearance of first examples of micromaps created in R. Three Web pages accompanying Carr & Pickle (2010) provide access to numerous data sets, shape files, and R code used in the book: <http://www.crcpress.com/product/isbn/9781420075731>, <http://www.statnetconsulting.com/micromaps.html>, and <http://mason.gmu.edu/~dcarr/Micromaps/>.

Example 1:

Execute this code to create a simplified micromap with one statistical panel, written by Mike Minnotte:

```
library(RColorBrewer)
library(maps)

display.brewer.pal(6, "Set1")
pal <- brewer.pal(6, "Set1")
pal[6] <- "#DDDDDD"

data(state)
murder <- state.x77[,5]
murder.name <- state.name[order(murder, decreasing = T)]
murder.name
murder <- sort(murder, decreasing = T)

pdf("Ch9_micromap_Ex1.pdf", width = 7.5, height = 10)

layout(matrix(1:36, nrow = 12, byrow = T), widths = c(1, 1, 2),
        heights = c(rep(4, 5), 1, rep(4, 5), 3))
#layout.show(36)

for (i in 1:10)
{
```

```

# compute colors, plot map (column 1)

par(mex = 0.5, mar = rep(.01,4))
if (i <= 5) m.col <- c(rep(pal[6], 25), rep(0, 25))
  else m.col <- c(rep(0, 25), rep(pal[6], 25))
m.col[(i-1)*5+1:5] <- pal[1:5]
map.m.col <- m.col[match.map("state", murder.name)]
map("state", fill = T, col = map.m.col, border = 0,
    xlim = c(-125, -65), ylim = c(25, 50))

# plot labels (column 2)

par(mar = rep(.1,4))
plot(0, 0, xlim = c(0, 1), ylim = c(0, 1), type = "n", bty = "n",
    xaxt = "n", yaxt = "n", xlab = "", ylab = "")
points(rep(.1, 5), seq(.9, .1, by = -.2), pch = 21, bg = pal[1:5], cex = 2)
text(rep(.18, 5), seq(.9, .1, by = -.2), murder.name[(i-1)*5+1:5], pos = 4, cex = 1.5)

# plot dotplot of values (column 3)

par(mar = rep(.1,4))
if (i == 10) plot(0, 0, xlim = range(murder), ylim = c(0, 1), type = "n",
    yaxt = "n", xlab = "", ylab = "")
else plot(0, 0, xlim = range(murder), ylim = c(0, 1), type = "n", xaxt = "n",
    yaxt = "n", xlab = "", ylab = "")
abline(h = seq(.9, .1, by = -.2), lty = 3, col = "grey")
points(murder[(i-1)*5+1:5], seq(.9, .1, by = -.2), pch = 21, bg = pal[1:5], cex = 2)

# separate states above and below median

if (i == 5) {for (j in 1:3){
plot(0, 0, xlim = c(0, 1), ylim = c(0, 1), type = "n", bty = "n",
    xaxt = "n", yaxt = "n", xlab = "", ylab = "")
abline(h = .5, lwd = 3, col = pal[6])
}}

```



```
# Plot through remaining (empty) cells

if (i == 10) for (j in 1:3)
  plot(0, 0, xlim = c(0, 1), ylim = c(0, 1), type = "n", bty = "n",
       xaxt = "n", yaxt = "n", xlab = "", ylab = "")
}

# Label for dot plots
text(0.5, 0.25, "Murders per 100K Population", cex = 1.5)

dev.off()
```

Example 2:

More sophisticated R code for micromaps, created by Dan Carr. The original version of this R code can be found at <http://classweb.gmu.edu/dcarr/eda/schedule.html> (Week 6). The R code posted here has been modified to run from our course Web page.

```
# Load R Functions

load(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Ch9_panellLayout.Rdata"))

# Load Data

stateUnemploy95 =
  read.csv(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Ch9_stateUnemployment95.csv"),
    row.names = 1, header = T)

stateNamesFips =
  read.csv(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Ch9_stateNamesFips.csv"),
    row.names = 1, header = T)

stateVBorders =
  read.csv(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Ch9_stateVisibilityBorders.csv"),
    row.names = NULL, header = T)

nationVBorders =
  read.csv(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Ch9_nationVisibilityBorders.csv"),
    blank.lines.skip = F, row.names = NULL, header = T)

# Create pdf (or jpg) Output

pdf("Ch9_micromap_Ex2.pdf", width = 7.5, height = 10)
#jpeg("Ch9_micromap_Ex2.jpg", width = 7.5, height = 10, units = "in", res = 72)

source(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat6560/RDataAndScripts/Ch9_linked_micromaps.R"))

dev.off()
```

5.3 Conditioned Micromaps

Carr & Pickle (2010), p. 7, state:

“The primary purpose of the conditioned micromap is data exploration, unlike the linked micromap plot, which is used most often for presentation. A major difference between these designs is that linked micromaps use a single ranked variable to partition the mapped regions into a linear sequence of panels. In contrast, conditioned micromaps use two ranked variables to partition the regions into a two-way grid of panels, with the rank order of one variable determining the row membership and the rank order of the second variable determining the column membership. Two slider bars set category cutpoints for these variables, allowing the analyst to dynamically explore the geographic patterns on the map, based on categorized values of the two auxiliary variables.”

In the literature, conditioned micromaps, also have been called Conditioned Choropleth Maps (CCMaps). Additional examples of CCMaps can be found in Carr, Wallin & Carr (2000), Carr et al. (2002), Carr et al. (2003), Carr et al. (2005), and Symanzik & Carr (2008).

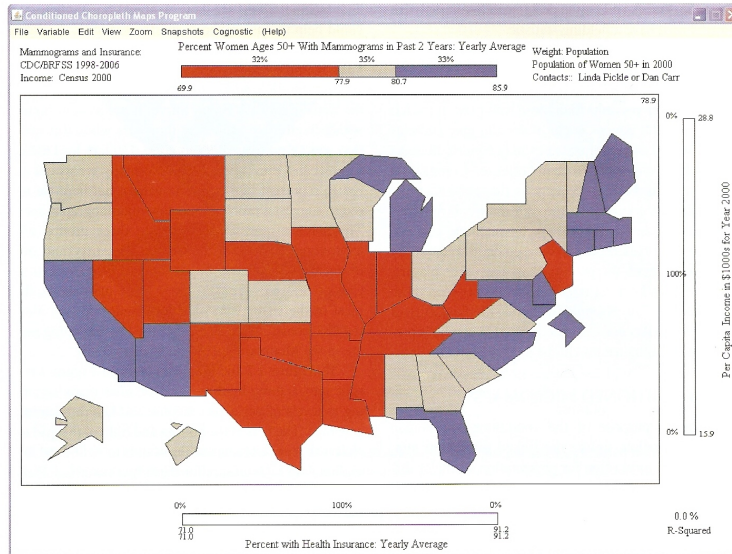


FIGURE 1.6 Full map of the percent of women ages 50 and over who have had a mammogram in the past two years, by state, which we partition into conditioned micromaps in Figure 1.7. Note that red is used as the low (bad) value, contrary to traditional use.

Figure 105: Carr & Pickle (2010), p. 8, Figure 1.6.

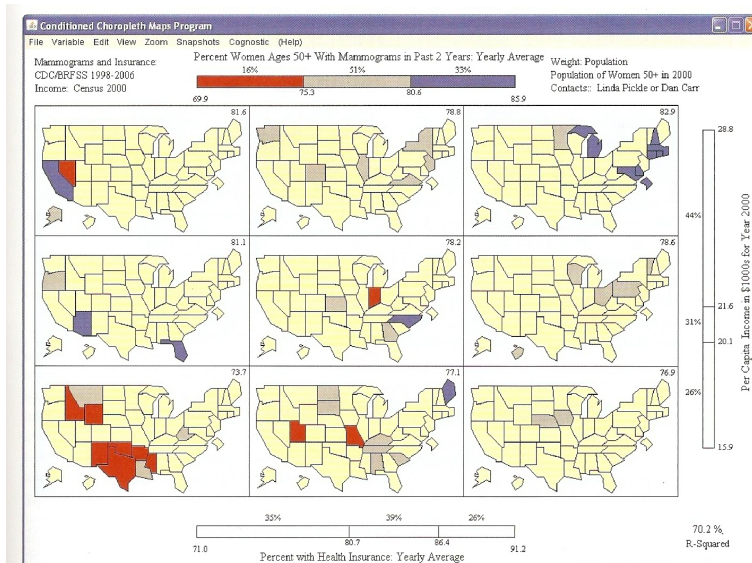


FIGURE 1.7 Conditioned micromap of the mammography data shown in Figure 1.6, conditioned on the per capita income (rows) and health insurance coverage (columns) as defined by the slider bars.

Figure 106: Carr & Pickle (2010), p. 9, Figure 1.7.

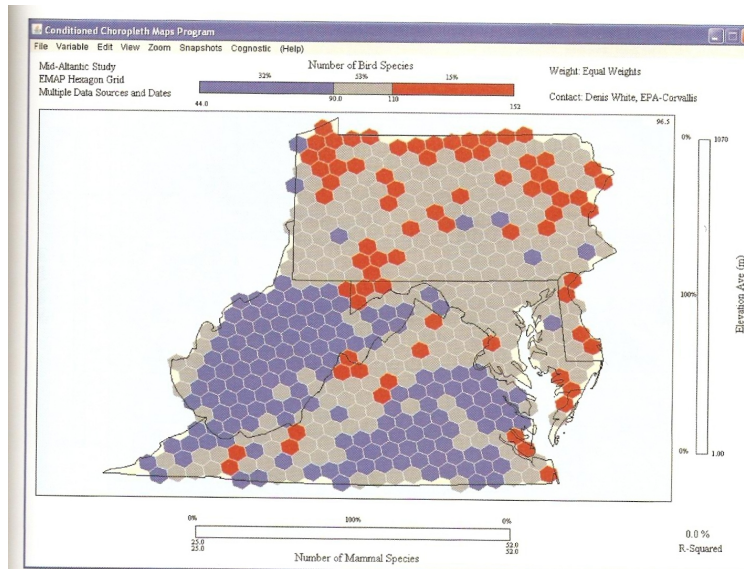


FIGURE 5.10 Use of a hexagon grid in conditioned micromaps to display the number of bird species in local regions. The blue hexagons have 90 or fewer bird species, the gray regions have 91 to 110 species, and the red hexagons have over 110 species (up to 152).

Figure 107: Carr & Pickle (2010), p. 91, Figure 5.10.

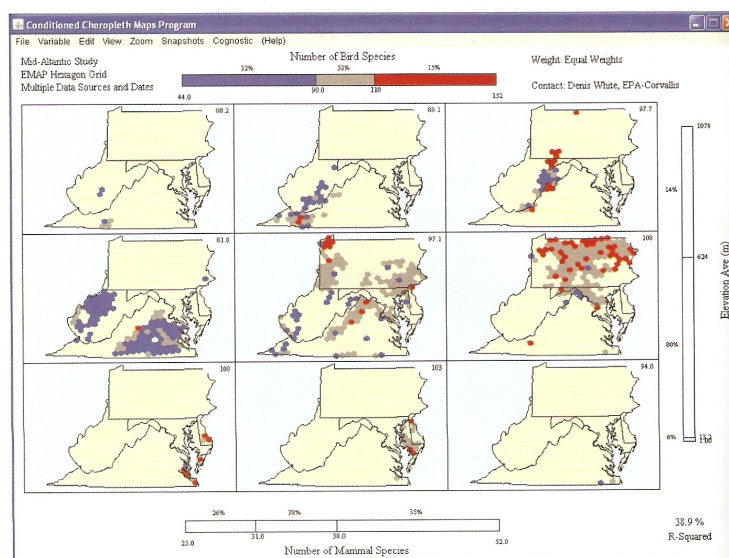


FIGURE 5.11 Hexagon bin conditioned choropleth micromap of bird species data (Figure 5.10) conditioned on the number of mammal species and average elevation.

Figure 108: Carr & Pickle (2010), p. 92, Figure 5.11.

5.4 Comparative Micromaps

Carr & Pickle (2010), p. 8, state:

“Comparative micromaps are one- and two-way sequences of maps indexed by time or other attributes. The emphasis is on comparisons across maps rather than comparisons within one map. The most common variant of comparative micromaps is a sequence of time-specific maps of values accompanied by a corresponding series of maps of the class or value differences of consecutive maps, allowing the reader to see explicitly the amount and location of changes. It is these difference maps that distinguish comparative micromaps from a standard time series of maps.”

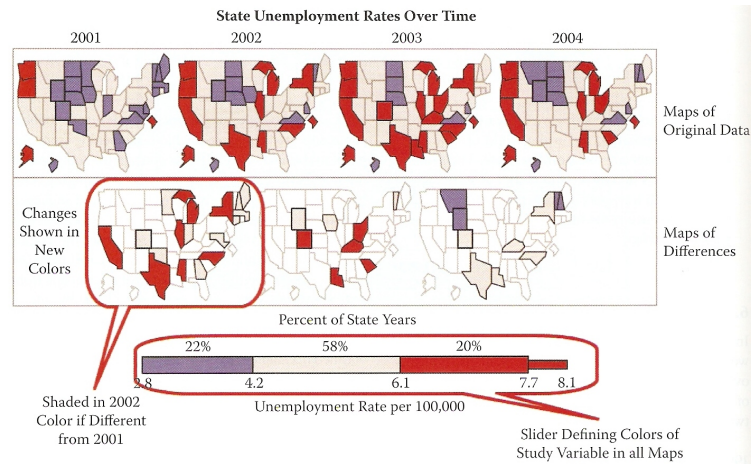


FIGURE 6.1 Comparative micromaps of annual state unemployment rates, 2001–2004, illustrating the basic design, annotated to show basic components.

Figure 109: Carr & Pickle (2010), p. 110, Figure 6.1.

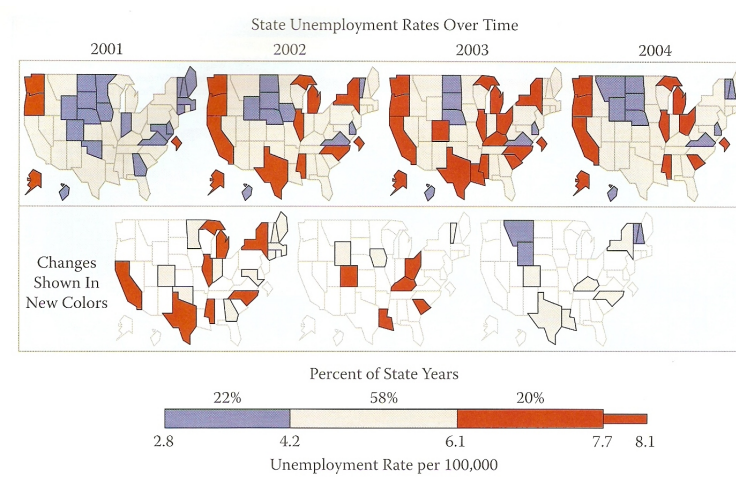


FIGURE 1.8 A typical comparative micromap plot design, with time series maps supplemented by explicit difference maps.

Figure 110: Carr & Pickle (2010), p. 10, Figure 1.8.

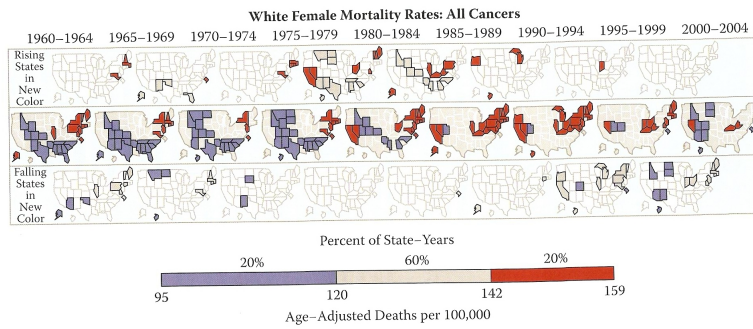


FIGURE 6.6 Rates of mortality among white females due to any malignancy, 1960-2004, by state, with changes shown separately for rising and falling rates.

Figure 111: Carr & Pickle (2010), p. 116, Figure 6.6.

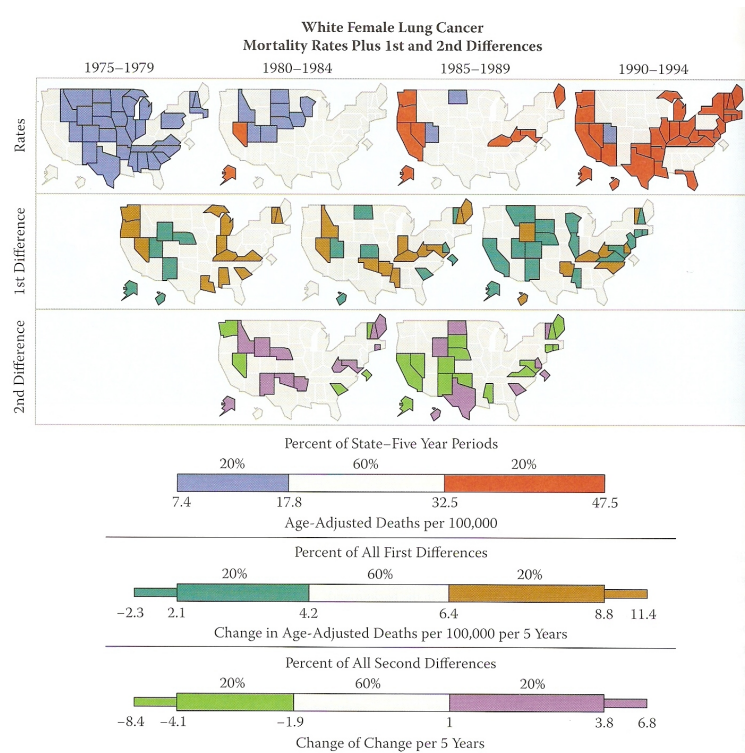


FIGURE 6.10 First differences (middle row) of the times series of white female lung cancer mortality rates by state (top row). Bottom row displays the second differences (differences of the second row).

Figure 112: Carr & Pickle (2010), p. 122, Figure 6.10.

5.5 Further Reading

Additional sources for choropleth maps and micromaps are:

- Work related to Pickle et al. (1996): Pickle & Herrmann (1999), Wainer (2008), and Pickle (2008)



Figure 113: http://www.cartoonstock.com/cartoonview.asp?start=4&search=main&catref=bven112&MA_Artist=&MA_Category=&ANDkeyword=statistics&ORkeyword=&TITLEkeyword=&NEGATIVEkeyword=, Cartoon.

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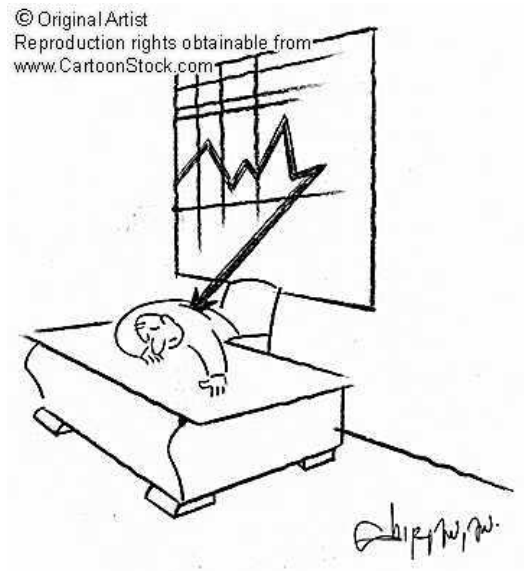


Figure 144: http://www.cartoonstock.com/blowup_stock.asp?imageref=vsh0184&artist=Shirvanian,+Vahan&topic=statistics+, Cartoon.